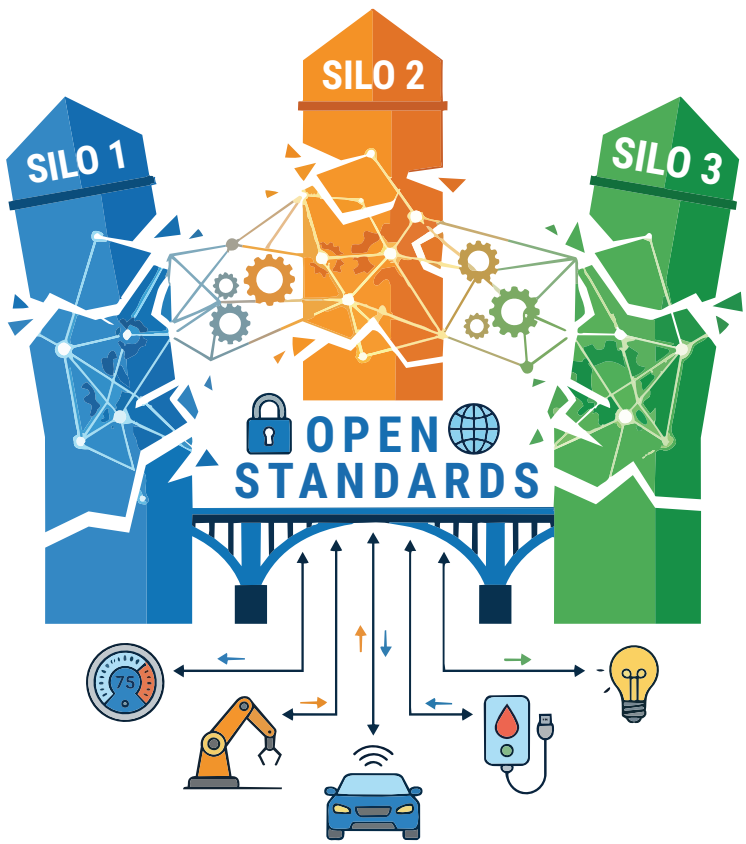
Breaking the Silos: How Open Standards Enable IoT Interoperability

To be truly interoperable on a large scale, IoT devices must communicate using open standards. Else, their utility will be restricted to small islands of connectivity.



The Internet of Things (IoT) introduced a great concept — devices connecting with and talking to each other. The smart meter communicating with your solar panel; the maintenance system communicating with a factory machine; and a sensor at the hospital communicating with the doctor's dashboard.

The truth is: it's not happening that way. Today, there are billions of connected devices. But many of them can't even comprehend each other. Each vendor uses its own radio, its own data format, and its own cloud. This turns into a world of small islands of connected devices, where data works well within but cannot be transferred between islands.

This is known as an interoperability problem — one of the most significant challenges to the success of IoT, making it costly, slow and difficult to scale up projects.

Fortunately, open standards are solving this problem. There are protocols, such as MQTT, CoAP and OPC UA, that provide devices with a common language. These standards are free and simple to use, thanks to open source tools.

What interoperability means

Interoperability is defined as the ability of different systems to interact with each other, giving each other information and applying it appropriately. This is done at four levels.

Technical level: Are the devices able to connect to each other? This is not about the local radio, networks and transport! Examples are Wi-Fi, Bluetooth, Zigbee, LoRaWAN and Ethernet.

Syntactic level: Will the devices be able to exchange messages? This is concerned with the format of messages and protocols. Examples are JSON, CBOR, MQTT and CoAP.

Semantic level: Do the devices "know" what the data means? One sensor can return 'temp: 25'. The other may send, 't: 77'. Both are temperatures. However, one is in Celsius and the other in Fahrenheit. This can be confusing since there is no meaning attached to the data.

Organisational level: Are there ways to share data safely on the basis of a mutual agreement between different companies and platforms? This includes security, access rules and business agreements.

Most IoT projects work fine at the first level. The true pain is experienced